

NEKHEL, Egypt

WMO: 624520

Lat: 29.9142N

Lon: 33.7402E

Elev: 1322

StdP: 14.01

Time Zone: 2.00 (E02)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	31.8	34.0	13.7	11.6	52.6	18.1	14.4	55.2	33.6	48.2	28.8	52.8	6.3	250	0.442

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	29.7	100.2	65.4	97.8	65.0	95.8	65.0	72.9	81.3	71.8	80.8	70.8	80.6	10.4	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
71.0	120.1	75.4	69.6	114.2	74.8	68.2	108.7	74.6	37.5	80.9	36.4	80.9	35.6	80.0	84.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
26.1	21.0	18.4	DB	27.3	105.7	3.8	2.5	24.6	107.5	22.4	108.9	20.2	110.3	17.5	112.1	(4)
			WB	24.2	77.0	3.1	3.9	21.9	79.7	20.1	82.0	18.3	84.1	16.0	86.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	66.5	49.6	53.4	58.6	65.0	72.1	77.1	80.5	80.7	76.7	70.4	60.3	52.2
	DBStd	11.94	4.58	5.96	6.41	6.20	5.56	3.82	2.74	2.53	3.51	4.79	5.64	4.40
	HDD50	116	61	25	3	0	0	0	0	0	0	0	1	26
	HDD65	1680	477	330	222	76	6	0	0	0	0	9	164	396
	CDD50	6124	49	119	269	449	684	814	947	952	802	633	311	95
	CDD65	2213	0	4	23	75	225	364	482	487	352	177	24	0
	CDH74	28034	16	106	476	1388	3201	4861	6373	5969	3738	1630	261	15
	CDH80	14483	1	26	174	631	1576	2694	3713	3357	1742	512	56	1
Wind	WSAvg	7.4	7.4	8.0	8.4	8.6	7.9	8.0	7.1	7.1	7.2	6.9	6.4	6.6
Precipitation	PrecAvg	1.1	0.3	0.2	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.1	0.1	0.2
	PrecMax	2.4	0.8	0.7	0.9	0.3	0.3	0.0	0.0	0.0	0.1	0.5	1.0	0.8
	PrecMin	0.4	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
	PrecStd	0.5	0.2	0.1	0.2	0.1	0.1	0.0	0.0	0.0	0.0	0.1	0.2	0.2
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	75.6	83.6	92.5	98.7	102.4	103.9	103.1	101.5	97.6	94.3	86.2	76.2
		MCWB	53.2	58.2	59.3	61.3	63.1	64.3	65.9	67.5	65.7	63.8	62.0	53.7
	2%	DB	69.7	76.8	84.7	92.5	97.2	98.8	99.5	98.2	94.1	88.9	80.7	70.2
		MCWB	50.3	53.7	57.1	58.9	62.2	64.6	65.8	66.7	66.6	64.9	60.3	52.4
	5%	DB	65.0	71.2	79.4	87.6	93.0	95.6	97.3	96.1	91.6	85.3	76.6	67.0
		MCWB	50.2	52.2	55.3	57.7	61.6	64.1	65.4	66.2	66.9	65.0	59.0	52.4
	10%	DB	61.5	66.5	73.6	82.0	89.1	93.0	95.1	94.0	89.0	82.2	73.1	64.2
		MCWB	49.2	50.7	53.4	56.4	60.9	64.1	65.6	66.5	66.9	64.7	58.1	51.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	55.8	60.0	61.6	63.4	66.6	70.9	74.0	74.7	75.3	71.8	66.7	58.2
		MCDB	67.7	76.0	82.2	88.4	83.3	79.5	81.5	83.1	82.6	81.3	74.6	66.6
	2%	WB	53.6	56.3	59.0	61.1	64.8	69.0	71.9	72.9	72.6	69.4	64.2	56.0
		MCDB	64.8	69.2	77.6	84.8	84.2	81.2	80.9	81.1	81.4	77.9	72.1	64.8
	5%	WB	51.6	54.0	56.6	59.4	63.6	67.7	70.6	71.9	71.1	68.3	62.4	54.2
		MCDB	62.3	66.6	75.1	80.9	84.0	82.5	80.5	81.1	79.8	76.8	71.9	64.1
	10%	WB	50.1	52.3	54.7	58.0	62.4	66.5	69.5	71.0	69.9	67.1	60.0	52.6
		MCDB	60.3	64.0	70.3	77.6	82.8	82.8	80.5	81.1	79.7	76.4	71.3	62.6
Mean Daily Temperature Range	5% DB	MDBR	24.9	25.4	27.2	29.8	31.1	31.2	29.7	27.9	25.9	24.7	26.0	25.6
		MCDBR	30.7	32.8	36.1	39.4	36.5	35.8	33.2	31.0	28.6	27.6	28.9	29.5
		MCWBR	17.5	16.4	16.0	15.9	13.0	12.0	10.3	9.6	9.5	10.6	13.6	17.0
	5% WB	MCDBR	26.1	26.9	31.3	32.6	31.8	30.7	28.1	27.1	24.3	23.2	23.8	25.3
		MCWBR	16.2	14.1	14.2	14.1	11.5	10.6	9.1	8.8	8.3	9.1	11.2	15.9
Clear-Sky Solar Irradiance	taub	0.346	0.381	0.432	0.481	0.488	0.408	0.408	0.408	0.415	0.423	0.361	0.340	
	taud	2.209	2.089	1.924	1.813	1.824	2.075	2.093	2.100	2.080	2.044	2.207	2.249	
	Ebn at Noon	277	277	269	260	259	279	278	277	271	259	269	273	
	Edh at Noon	38	47	59	68	68	52	51	50	50	48	37	35	
All-Sky Solar Radiation	RadAvg	1166	1457	1878	2219	2456	2660	2579	2386	2069	1650	1298	1094	
	RadStd	44	52	69	55	70	26	39	33	41	50	35	39	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (12 neighbors)	+0.72	N/A	-1.91	N/A	N/A	N/A	N/A	N/A	+243	+263	

Nomenclature: See separate page